

FREDRIKA, Sweden

WMO: 022630

Lat: 64.0744N

Lon: 18.3619E

Elev: 327

StdP: 97.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.1	-21.8	-27.5	0.3	-25.1	-23.9	0.4	-21.7	9.9	-2.6	6.7	-3.5	0.7	350	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	24.9	15.9	22.9	15.4	21.0	14.5	17.6	21.9	16.5	20.8	15.5	19.4	2.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.0	11.9	19.0	14.9	11.0	17.9	13.8	10.3	16.7	50.8	21.9	47.5	20.8	44.4	19.5	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
6.0	5.2	4.5	-29.4	27.1	3.2	2.2	-31.7	28.7	-33.5	29.9	-35.3	31.1	-37.6	32.7		
			-29.4	18.9	3.0	1.3	-31.6	19.9	-33.3	20.6	-35.0	21.3	-37.2	22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-8.9	-8.4	-4.1	1.5	7.3	11.8	15.0	13.1	8.2	2.3	-3.0	-6.4	(6)
(7)		DBStd	9.40	6.39	6.44	4.95	3.54	3.95	3.38	3.05	3.02	3.13	3.77	4.79	6.08	(7)
(8)		HDD10.0	3128	585	515	438	254	104	18	1	7	69	238	391	508	(8)
(9)		HDD18.3	5811	843	748	696	504	343	197	111	163	303	496	641	766	(9)
(10)		CDD10.0	370	0	0	0	0	20	73	156	104	17	1	0	0	(10)
(11)		CDD18.3	12	0	0	0	0	0	2	8	2	0	0	0	0	(11)
(12)		CDH23.3	137	0	0	0	0	3	32	82	19	0	0	0	0	(12)
(13)		CDH26.7	14	0	0	0	0	0	2	11	1	0	0	0	0	(13)
(14)	Wind	WSAvg	2.1	1.9	1.9	2.3	2.3	2.4	2.4	2.1	1.9	2.0	2.1	1.9	1.8	(14)
(15)	Precipitation	PrecAvg	598	39	28	28	31	40	61	92	81	58	52	48	42	(15)
(16)		PrecMax	826	69	69	65	57	72	104	218	171	140	131	97	86	(16)
(17)		PrecMin	438	3	4	9	7	7	24	16	18	19	20	23	5	(17)
(18)		PrecStd	96	15	16	13	16	18	21	43	44	31	25	24	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	5.2	8.6	17.0	23.7	26.6	28.2	25.6	18.7	12.3	8.2	4.6	(19)
(20)		0.4%	MCWB	3.1	3.1	4.6	8.5	13.1	14.9	17.6	17.0	13.8	9.4	7.4	3.5	(20)
(21)		2%	DB	2.5	3.6	6.0	12.5	20.6	23.7	25.5	23.1	16.6	10.3	5.9	3.1	(21)
(22)		2%	MCWB	1.5	1.9	2.7	6.7	12.2	14.3	16.7	16.2	12.7	8.4	5.1	2.1	(22)
(23)		5%	DB	0.8	1.6	4.5	10.0	18.1	20.7	23.5	21.1	14.7	9.0	4.1	1.8	(23)
(24)		5%	MCWB	0.2	0.4	1.8	4.9	10.9	13.2	15.8	15.5	11.6	7.7	3.5	1.2	(24)
(25)		10%	DB	-0.6	0.1	3.0	7.8	15.2	18.6	21.6	18.9	13.4	7.8	2.7	0.7	(25)
(26)		10%	MCWB	-1.0	-0.7	0.9	3.8	9.3	12.0	15.3	14.7	10.9	6.8	2.3	0.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.3	4.7	9.0	14.6	17.0	19.7	18.7	14.8	10.3	7.5	3.7	(27)
(28)		0.4%	MCDWB	4.4	4.8	7.9	16.1	20.8	23.6	24.8	22.9	17.1	11.4	8.2	4.3	(28)
(29)		2%	WB	1.6	2.0	3.2	7.0	12.9	15.2	18.1	17.2	13.4	8.9	5.3	2.2	(29)
(30)		2%	MCDWB	2.3	3.3	5.5	12.0	19.4	20.9	22.1	20.9	15.4	9.8	5.8	2.8	(30)
(31)		5%	WB	0.2	0.6	2.1	5.4	11.3	14.1	17.1	16.1	12.4	7.9	3.6	1.2	(31)
(32)		5%	MCDWB	0.6	1.4	4.1	9.0	17.3	19.3	21.8	19.6	14.1	8.6	4.1	1.6	(32)
(33)		10%	WB	-1.1	-0.7	1.0	4.1	9.6	13.0	16.1	15.0	11.4	6.8	2.3	0.3	(33)
(34)		10%	MCDWB	-0.7	-0.2	2.9	7.4	14.5	17.1	20.2	18.1	12.8	7.8	2.6	0.6	(34)
(35)	Mean Daily Temperature Range	MDBR	6.4	7.2	8.5	8.7	9.8	9.4	9.2	8.5	7.6	5.7	4.4	6.0	(35)	
(36)		5% DB	MCDBR	6.3	6.4	9.0	12.8	14.7	13.1	12.5	12.4	9.7	7.1	4.7	5.6	(36)
(37)		5% DB	MCWBR	6.1	5.5	6.4	7.4	7.3	5.6	5.2	5.8	6.0	5.4	4.4	5.4	(37)
(38)		5% WB	MCDBR	6.2	5.8	8.6	11.5	14.1	11.4	10.8	10.2	8.4	6.5	4.5	5.4	(38)
(39)		5% WB	MCWBR	5.9	5.2	6.4	7.1	7.3	5.4	5.2	5.4	5.8	5.4	4.3	5.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.199	0.266	0.294	0.330	0.334	0.334	0.347	0.335	0.306	0.290	0.348	0.350	(40)	
(41)		taud	1.997	2.199	2.267	2.345	2.469	2.496	2.471	2.496	2.566	2.448	2.749	2.740	(41)	
(42)		Ebn at Noon	388	656	797	848	880	886	861	839	800	643	369	208	(42)	
(43)		Edh at Noon	26	57	82	97	94	94	93	83	63	46	23	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.15	0.76	2.26	3.87	4.80	5.38	4.95	3.75	2.26	0.92	0.23	0.06	(44)	
(45)		RadStd	0.02	0.13	0.22	0.31	0.57	0.41	0.54	0.35	0.27	0.16	0.05	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-133	N/A	N/A	N/A	(47)

Nomenclature: See separate page